

Simple 12V, 1A SMPS

ASHVINI VISHVAKARMA AND
ATANU DASGUPTA

Most electronics enthusiasts require DC power supplies to operate various devices and accessories. The most popular and common supply is a 12V DC supply that can be easily derived from the household AC supply with transformation, rectification, filtering and stabilisation. These power supplies have a bulky steel- or iron-laminated transformer that provides a safety barrier for the low-voltage output from the AC input, and reduces the input from

typically 230V AC to a much lower voltage. The low-voltage AC output from the transformer is then rectified by two or four diodes and smoothed into low-voltage DC by large electrolytic capacitors.

A switched mode power supply (SMPS) offers the same end results at a lower cost and higher efficiency. For a given output power, an SMPS is lighter and smaller. This is because, if the frequency of operation is increased, one can get away with using a smaller core cross-sectional area. Besides, an iron-core transformer works only up to about 10 kHz, and if we

need something in 50-100kHz range, we need a ferrite core.

Circuit and working

Fig. 2 shows the circuit of a simple 12V, 1A SMPS. The circuit is built

around a low-power offline switcher TNY266 (IC1), photo-transistor photocoupler EL817 (IC2), a flyback transformer (X1) and some other easily-available components.

Low-power offline switcher (TNY266). The SMPS here has been designed using a TNY266 chip, which is affectionately called the '555' of SMPS. This device has a 700V power MOSFET, an oscillator, a high-voltage switched current source, a current limiting and thermal shutdown circuitry integrated onto a monolithic device. The start-up and operating power is derived directly from the voltage on the drain (pin 5), eliminating the need for a bias winding and associated circuitry. In addition, the device incorporates auto-restart, line under-voltage sense, and frequency jittering.

The drain-source breakdown voltage of the MOSFET in TNY266 is important. During the 'off' period, the MOSFET sees rectified 317V DC approximately. Additionally, it sees the reflected voltage of the secondary, which is about 130V AC. It also encounters the ringing voltage from the leakage inductance, and the drain-source capacitance of the MOSFET. Therefore a MOSFET with a V_{DS} of 650V DC is expected to keep the necessary safe operating margin. Fortunately, a MOSFET with these properties is included in TNY266.

The 230V AC input is connected at CON1, which is rectified by diode D1. The neon lamp (NL1) glows when the input supply is present. Resistor R1 limits the current through the lamp. The rectified output goes to the first terminal (A) of coil L1 and the second terminal (B) is connected to the drain of the inbuilt MOSFET in IC1.



Fig. 1: The author's prototype

TABLE I
Computation of Winding Details for the SMPS

Specifications	Details
Minimum AC mains input	200V
Secondary output voltage/current	12V/1A DC
Output power [P_o]	12W
Peak primary voltage post rectification [$V_{\text{primary.pk}}$]	$200 \times \sqrt{2} = 282V$
Voltage drop in primary [$V_{\text{mains.drop}}$] ($D1 \text{ drop} + R2 \text{ drop} + AC \text{ ripple}$)	$10\% \text{ of } V_{\text{primary.pk}} = 28V$
Operating frequency	120kHz; Tiny switch-II family
Cycle time	$1/120kHz = 8.3 \mu s$
Max duty cycle [DC]	0.45
Primary on time [$T_{\text{on.primary}}$]	$0.45 \times 8.3 \mu s = 3.7 \mu s$
Secondary on time [$T_{\text{on.secondary}}$]	$(1-0.45) \times 8.3 \mu s = 4.6 \mu s$
Flux [Φ] = $B_{\text{sat}} \times \text{Core area}$	$0.24T \times (25 \times 10^{-6})m^2 = 6 \times 10^{-6} Wb$
Primary turns = $[(V_{\text{primary.pk}} - V_{\text{mains.drop}}) \times T_{\text{on.primary}}] / \Phi$	157 turns
Secondary turns = $1.5 \times (V_{\text{secondary}} \times T_{\text{on.secondary}}) / \Phi$	14 turns
Peak primary current [$I_{\text{primary.pk}}$] = $(3 \times P_o) / [(DC \times (V_{\text{primary.pk}} - V_{\text{mains.drop}}))]$	0.33A
Peak secondary current = $(3 \times P_o) / [(1-DC) \times V_{\text{secondary}}]$	5.4A